Physics-Informed LLM Latency Prediction:
Critical Analysis of the Analytical Roofline Model vs. Direct Operational Modeling

Pre-Execution Scheduling & The Token Count Prerequisite

In production LLM serving, latency predictors must operate **strictly pre-execution** using launch configurations: cluster parallelism (`tp`, `pp`, `dp`, `num_gpus`) and request budgets (`batch_size`, `output_tokens`). Post-execution telemetry is strictly excluded.

The Token Prerequisite (Imai et al., Section A.4): The analytical Roofline Model (RLM) calculates hardware ceilings from input and output tokens:

$$T_{\text{roofline}} = T_{\text{prefill}}(\text{input_tokens}) + \text{output_tokens} \times T_{\text{decode_step}}(\text{batch_size}) \tag{1}$$

While prompt length is known pre-execution, output length is dynamic. As acknowledged by Imai et al. (Appendix A.4): *"In practical application, the generation length is not predetermined... To accurately model inference latency in real world scenarios, we must consider predicting the generation length based on the input prompt."*

Methodological Analysis: Operational Modeling vs. Analytical Physics:

- **Data Aggregation**: To prevent duplicate data leakage across repeated benchmark trials, the 10 replicate runs per setting are aggregated into unique configuration means ($N = 24$ unique configurations for Qwen3-14B) prior to modeling.
- **Feature Collision (Why Base Fails)**: Without a workload indicator, identical launch configurations (`batch_size=128, out=500, DP2`) predict both Alpaca (≈ 54 ms) and Longbench (≈ 452 ms). This severe collision causes baseline R^2 to collapse ($R^2 < 0$).
- **Resolving Workloads via Categorical Task**: Providing a categorical task indicator (`task: Alpaca vs. Longbench`) resolves the collision, allowing tree models to achieve $R^2 \approx 0.63$.
- **Value of Analytical Roofline**: The analytical Roofline Model provides a physical inductive bias that significantly benefits linear estimators ($R^2 = 0.2791 \rightarrow 0.5024$, 42% MSE reduction), while providing robust out-of-distribution generalization across unseen parallelism configurations (GroupKFold $R^2 = 0.44 - 0.51$).

Table 1: Qwen3-14B: Feature Set Ablation on Unique Configuration Means (R^2 / MSE).
Evaluated on 24 aggregated configuration cells via 3-Fold Cross-Validation. Base features: `[tp, pp, dp, num_gpus, batch_size, output_tokens]`.

Model / Estimator	Base (Collision)	Base + Categorical Task	Base + Analytical RLM	Key Takeaway
Linear Regression (LR)	-0.9465 / 373347	0.2791 / 140298	**0.5024 / 80841**	RLM provides linear prior
Random Forest (RF)	-0.7947 / 397206	**0.6265 / 100881**	0.5779 / 82980	Resolves workload collision
Gradient Boosting (GB)	-1.5466 / 522026	0.5644 / 109313	**0.6451 / 107853**	Strong tree performance
Out-of-Distribution Generalization (+Roofline, GroupKFold across Unseen Parallelism)				
Linear Regression: $R^2 = 0.4370$ (MSE 109206); Random Forest: $R^2 = 0.5136$ (MSE 109608); Gradient Boosting: $R^2 = 0.4812$ (MSE 139475)				

Key Conclusions for Deployment & Next Steps

- **Unique Configuration Modeling**: Aggregating repeated physical trials ensures clean, leakage-free evaluation without artificial lookup-table memorization.
- **Workload Characterization**: Static token assignments are replaced with explicit workload identifiers. Production deployments should capture continuous prompt length distributions from client traffic traces.
- **Physics as an Inductive Prior**: Analytical Roofline latency serves as a vital physical anchor for linear estimators ($R^2 = 0.2791 \rightarrow 0.5024$) and cross-architecture hardware scaling.
- **Experimental Next Steps**: Scale evaluation across heterogeneous architectures (14B to 27B) on BSC MareNostrum 5 under continuous load distributions.